

CNT4007 Project 2

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Overview

Given the instructions, it was clear TCP was the protocol to be chosen because we want a connection between the client and server, and we want reliability as a priority over speed, which TCP provides.

To handle multiple concurrent client requests, multithreading using pthreads was implemented so that the server can handle these multiple requests efficiently without having clients wait for slow tasks being processed for other clients.

To implement the "diff" between the server and the client, the CRC32 checksum algorithm was used to compare file contents without sending the entire file. This allows the server to compute a compact and efficient checksum for each file and send it to the client. The client, in turn, computes the checksum of its local files and compares it to the server's checksum to determine whether the files are identical, different, or missing. This ensures that only the necessary files are sent to the client during the "Pull" operation, avoiding unnecessary data transfer and meeting the project requirement of not sending the entire file for content comparison.

Functionality

Once a connection is established between a server and a client, a pthread is created to handle that specific connection. This ensures that multiple clients can be serviced concurrently without blocking other client operations.

In the connection-handling function, the client sends a message to the server, which responds with a confirmation of connection. The confirmation message also includes the list of possible inputs that the client can choose from:

List - Lists all the files on the server.

Diff - Compares the client's files to the server's using CRC32 checksums. The server sends the checksums for each file to the client, which compares them with its local files' checksums. If a file is missing or has different content, it is flagged for the client.

Pull - After performing the Diff, the client can choose to Pull the missing or different files from the server. The server then sends only those files, ensuring that unnecessary file transfers are avoided.

Leave - Ends the connection session between the client and server.

If 4 (Leave) is selected, the client sends the input to the server, which responds with a disconnect message. The server then closes the connection for that client. Once the client receives the disconnect message, it also closes its connection, and the session is terminated.

Architecture Design Summary

The project uses a client-server architecture over TCP.

- Server:
 - Listens for client connections and uses multithreading to handle multiple clients concurrently.
 - Performs file listing, comparison (Diff), and transfer (Pull), using CRC32 checksums to compare files without transferring their contents unless requested.
- Client:
 - Connects to the server, requests file listings, compares files using checksums, and pulls missing files as needed.
 - Computes local CRC32 checksums and compares them with those from the server during the Diff operation.
- Communication:
 - Communication is handled via TCP sockets in a request-response pattern. The client sends requests, and the server responds with the corresponding data or action.